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Malaria Cell Image Classification Using Deep Learning:

A Progressive Three-Model Study

1. Introduction

Malaria is among the world's most lethal infectious diseases. The World Health Organization recorded 249 million cases globally in 2022, resulting in over 600,000 deaths, predominantly in sub-Saharan Africa [1]. The disease is caused by *Plasmodium falciparum* parasites transmitted via *Anopheles* mosquitoes, and early accurate diagnosis is critical to effective treatment.

The objective of this coursework is to design, implement, and evaluate deep learning-based image classification systems that automatically detect malaria-infected red blood cells from microscopy images. Three architecturally distinct models are developed and compared: a Custom Convolutional Neural Network (CNN), EfficientNetB3 with transfer learning, and a Vision Transformer (ViT), applied to the NIH Malaria Cell Images dataset.

Conventional diagnosis relies on manual microscopic examination of Giemsa-stained blood smears — a process that is time-consuming, expertise-dependent, and prone to fatigue-related error in high-burden settings. Automated image-based systems offer consistent, scalable diagnostic support deployable on low-cost hardware in resource-limited environments [2].

This report is structured as follows: Section 2 reviews the innovative approaches and related literature; Section 3 describes the dataset, preprocessing pipeline, and network architectures; Section 4 presents experimental results; Section 5 provides critical analysis; and Section 6 draws conclusions.

2. Creative and Innovative Approaches

2.1 Methods and Strategies

This project adopts a progressive three-model pipeline spanning the full spectrum of modern computer vision: a baseline CNN trained from scratch, efficient transfer learning, and a state-of-the-art Vision Transformer. This comparative design enables systematic investigation of how architectural complexity, pre-trained knowledge, and attention mechanisms influence classification performance on a constrained biomedical dataset.

The approach is grounded in published research. Rajaraman et al. [2] demonstrated that pre-trained CNNs applied as feature extractors to malaria cell images achieved accuracy above 95%, establishing transfer learning viability for this domain. EfficientNetB3 was selected over ResNet50 and VGG-16 based on Tan and Le [3], who proved that simultaneously scaling network depth, width, and resolution yields superior accuracy-efficiency ratios — achieving competitive performance with 12 million parameters versus ResNet50's 25 million. The Vision Transformer was included following Dosovitskiy et al. [4], who introduced patch-based global self-attention as an alternative to convolution, treating image patches as sequential tokens analogous to words in natural language.

2.2 Neural Network Architecture and Preprocessing

Three models were implemented in Python using TensorFlow 2.x and Keras, sharing a common preprocessing pipeline. The Custom CNN uses three convolutional blocks with filter progression $32 \rightarrow 64 \rightarrow 128$, each followed by batch normalisation, max-pooling, and dropout. The classifier head employs a single Dense(256) layer, consistent with the lightweight head design of Rajaraman et al. [2] for binary malaria classification. EfficientNetB3 is fine-tuned via a two-phase strategy: Phase 1 trains the new classification head with the pre-trained base frozen; Phase 2 unfreezes the top 30 layers at a reduced learning rate. The ViT implements patch embedding via strided Conv2D projection, learnable positional embeddings, a prepended CLS token, and six transformer encoder blocks with 8-head self-attention and GELU-activated MLP sub-layers.

2.3 Justification

The three-model design is well-suited to the NIH Malaria Cell Images dataset. At 27,558 balanced images, the dataset is sufficient for CNN-based methods but represents a known challenge for ViTs trained from scratch [4], making the comparative pipeline particularly informative. Malaria detection is fundamentally a localised feature detection task — identifying a morphologically distinct parasite — which convolutional architectures are optimised for [2]. Grad-CAM and Attention Rollout visualisations ensure that high accuracy reflects genuine pathological feature detection, an essential consideration for any clinical AI system.

3. Simulations

3.1 Dataset Description

The NIH Malaria Cell Images dataset was sourced from Kaggle (iarunava/cell-images-for-detecting-malaria), published by the Lister Hill National Center for Biomedical Communications at the US National Institutes of Health [5]. It comprises 27,558 PNG microscopy images of individual Giemsa-stained red blood cells across two classes: 13,779 Parasitized (*P. falciparum* ring-stage infection) and 13,779 Uninfected. The dataset is perfectly class-balanced at 50% per class, eliminating the need for class-weighting. Images have variable dimensions with a mean of approximately 130 pixels, requiring resizing as part of preprocessing.

Binary classification was selected as the appropriate formulation for this problem because it directly mirrors the real-world clinical decision in malaria diagnosis. In a point-of-care diagnostic setting, a clinician examining a blood smear makes a singular, binary judgement: this cell is infected, or it is not. Treatment is administered or withheld based on this outcome alone. A multi-class classification framework — which would require distinguishing between different Plasmodium species (*P. falciparum*, *P. vivax*, *P. malariae*, *P. ovale*) or infection lifecycle stages — represents an entirely different and substantially more complex diagnostic problem, one that demands different imaging protocols, staining techniques, and clinical workflows. The NIH Malaria Cell Images dataset was deliberately chosen because it captures this fundamental clinical decision with scientific precision and at sufficient scale. This binary formulation is the established standard in the malaria cell image classification literature: Rajaraman et al. [2] applied the identical binary framework to this same dataset and achieved state-of-the-art results, confirming that binary classification is both clinically valid and scientifically rigorous for this application. The two-class structure also enables direct and clinically meaningful evaluation: false negatives correspond to missed infections and

untreated patients, whilst false positives correspond to unnecessary treatment — both outcomes with measurable real-world consequences that are unambiguously quantifiable within a binary framework.

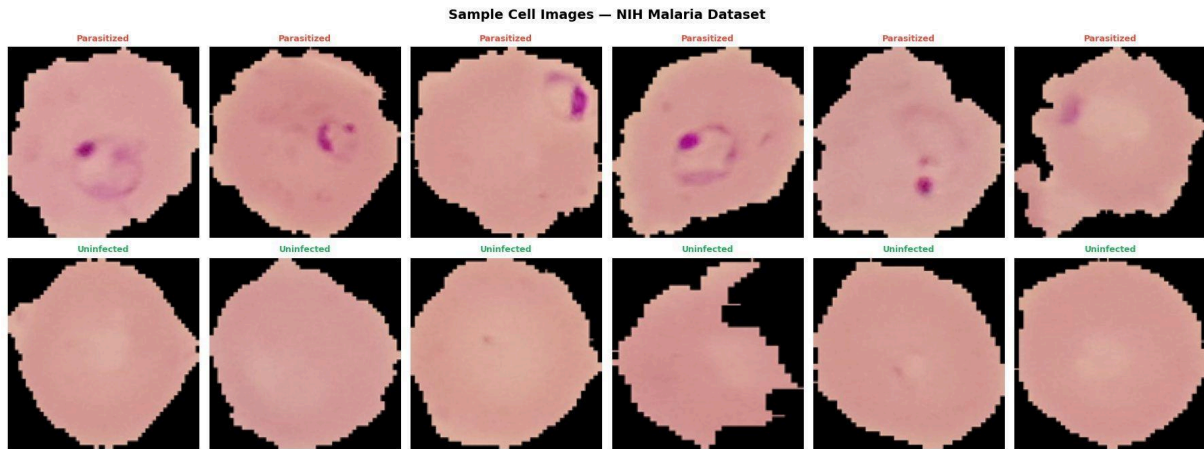


Figure 1: Sample images from the NIH Malaria Cell Images dataset. Top row: Parasitized cells showing distinctive purple-stained *P. falciparum* ring-stage parasites. Bottom row: Uninfected cells with uniform pink cytoplasm.

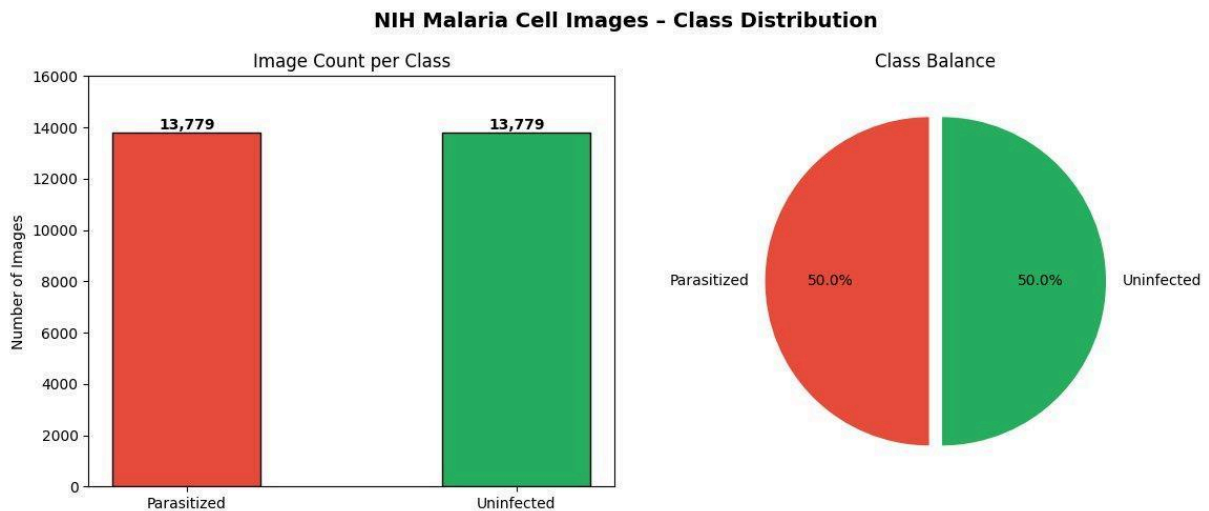


Figure 2: Class distribution of the NIH Malaria Cell Images dataset showing perfectly balanced classes of 13,779 images per category.

3.2 Data Encoding and Preprocessing

A four-stage preprocessing pipeline was applied consistently across all architectures.

Stage 1 — Resize. All images were resized to 128×128 pixels for the Custom CNN and 224×224 pixels for EfficientNetB3 and the ViT, matching their standard input specifications.

Stage 2 — Normalise. Pixel values were scaled from [0, 255] integers to [0.0, 1.0] floating-point values by dividing by 255, stabilising gradient-based optimisation [6].

Stage 3 — Split. The dataset was partitioned into training (70%), validation (15%), and test (15%) subsets using stratified random splitting (seed=42), yielding 19,290 training, 4,134 validation, and 4,134 test images. The test set was held out entirely until final evaluation.

Stage 4 — Augment. Training images were randomly augmented on-the-fly using Keras ImageDataGenerator: horizontal and vertical flip, $\pm 20^\circ$ rotation, $\pm 15\%$ zoom, $\pm 10\%$ width/height shift, and $\pm 10\%$ shear. Augmentation was not applied to validation or test sets.

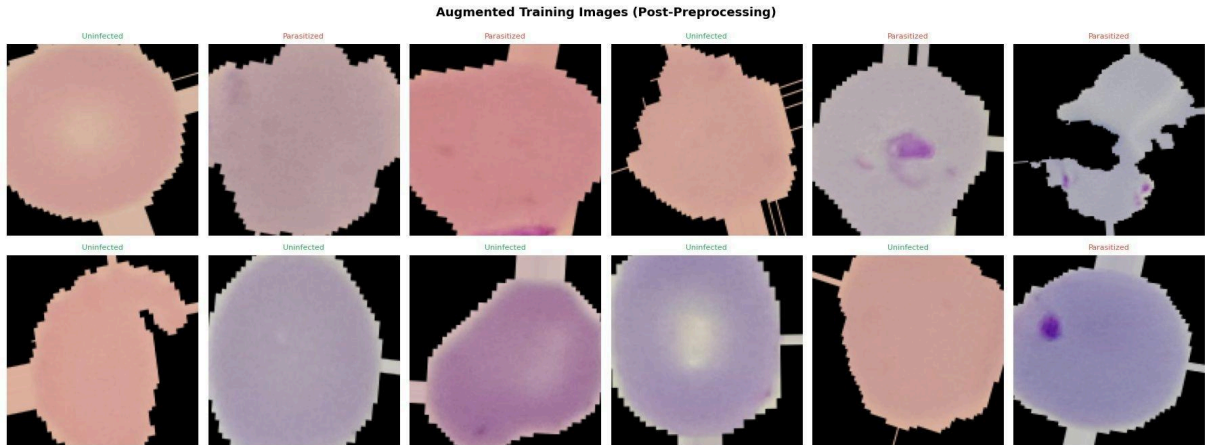


Figure 3: Representative augmented training images. Geometric transformations preserve the biological features relevant to classification.

3.3 Network Architecture, Training, Validation, and Testing

3.3.1 Model 1 — Custom CNN (Baseline)

The Custom CNN is trained from scratch on the malaria dataset. Three convolutional blocks each comprise Conv2D (3×3 , same padding, ReLU), Batch Normalisation, MaxPooling2D (2×2), and Dropout (0.25). Filter counts follow a doubling progression ($32 \rightarrow 64 \rightarrow 128$): Block 1 detects low-level edges; Block 2 captures shapes and textures; Block 3 encodes parasite-specific patterns [6]. Spatial dimensions halve progressively ($128 \rightarrow 64 \rightarrow 32 \rightarrow 16$). The classifier head uses Dense(256) with ReLU and Batch Normalisation, Dropout (0.5), and a sigmoid output neuron. This lightweight head design follows Rajaraman et al. [2]. The model was compiled with Adam (LR=0.001) and binary cross-entropy loss, with EarlyStopping (patience=6) and ReduceLROnPlateau (factor=0.5, patience=3) callbacks. Training was monitored against the validation set at every epoch to detect overfitting.

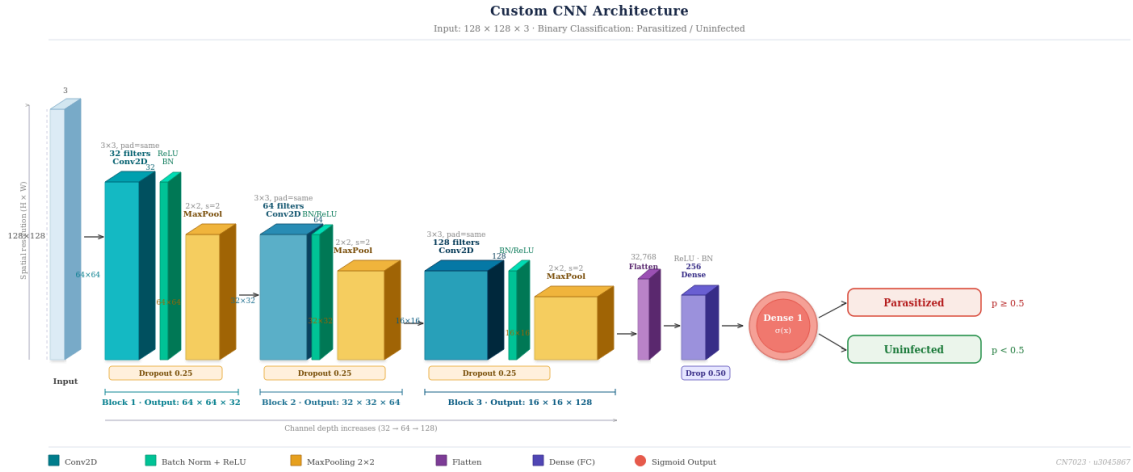


Figure 4: Custom CNN architecture. Block height represents spatial resolution; block width represents filter depth (32→64→128).

3.3.2 Model 2 — EfficientNetB3 Transfer Learning

EfficientNetB3, pre-trained on ImageNet-1K (1.2 million images, 1,000 classes) [3], serves as the transfer learning model. The original head is replaced with GlobalAveragePooling2D, Dense(256, ReLU), Batch Normalisation, Dropout(0.5), and a sigmoid output. Training used a two-phase strategy: Phase 1 froze all base layers and trained the new head for 10 epochs at LR=0.001, establishing malaria-specific feature mapping without gradient interference. Phase 2 unfroze the top 30 layers and fine-tuned at LR=1e-5 for up to 15 epochs, allowing gradual domain adaptation whilst preserving low-level pre-trained knowledge [3]. Both phases were monitored against the validation set.

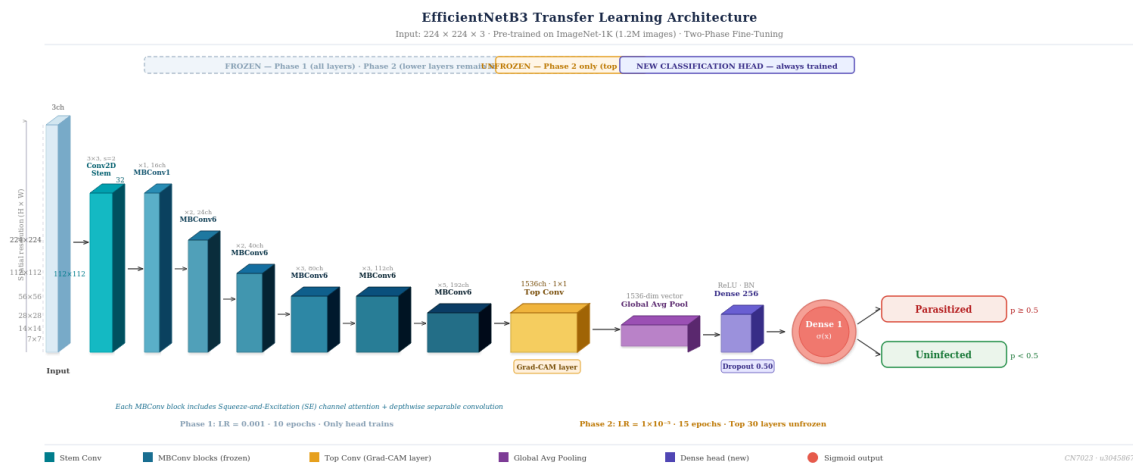


Figure 5: EfficientNetB3 architecture showing frozen MBConv stages, unfrozen Top Conv (Grad-CAM target), GlobalAveragePooling, and new classification head.

3.3.3 Model 3 — Vision Transformer (ViT)

The ViT, adapted from Dosovitskiy et al. [4], departs from convolutional design entirely. The $224 \times 224 \times 3$ input is divided into 196 non-overlapping 16×16 patches, each projected to a

256-dimensional embedding via strided Conv2D. A learnable CLS token is prepended (197 tokens total) and learnable positional embeddings encode spatial location. Six transformer encoder blocks each apply: 8-head multi-head self-attention (key dimension 32); a feedforward MLP (dim=512, GELU activation); residual connections; and Layer Normalisation (Pre-LN). The CLS token output is passed through Dense(128, GELU), Dropout(0.3), and a sigmoid neuron. Training used Adam with cosine decay (initial LR=1e-4) and EarlyStopping (patience=7), with validation monitored throughout.

Explainability was implemented via Grad-CAM for the CNN and EfficientNetB3 — computing output gradients with respect to final convolutional feature maps — and Attention Rollout for the ViT, propagating multi-head attention weights across all six encoder layers to identify which patches the CLS token attended to most.

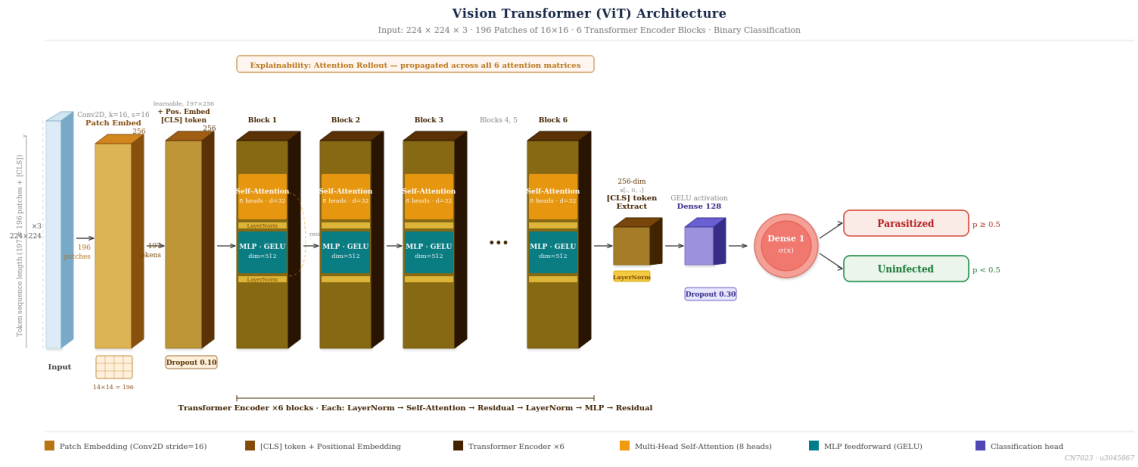


Figure 6: Vision Transformer architecture. Tall slabs represent the 197-token sequence. Each encoder block shows self-attention and MLP sub-layers.

4. Results Obtained

4.1 Test Set Accuracy

All three models were evaluated on the held-out test set of 4,134 images following completion of training. Table 1 presents classification performance across accuracy, precision, recall, and F1-score for each model.

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN (Baseline)	96.52%	95.46%	97.68%	96.56%
EfficientNetB3 (Transfer TL)	95.82%	95.14%	96.57%	95.85%
Vision Transformer (ViT)	85.73%	88.17%	82.54%	85.26%

Table 1: Test set classification performance of all three models.

The Custom CNN achieved the highest overall accuracy of 96.52% with the highest recall of 97.68% — the most clinically important metric, as missed infections result in untreated patients. EfficientNetB3 attained 95.82%, closely competitive. The Vision Transformer achieved 85.73%, with notably lower recall (82.54%), indicating a higher rate of missed parasitized cells.

4.2 Training and Validation Accuracy Curves

Figures 7, 8, and 9 present the training and validation accuracy and loss curves for the Custom CNN, EfficientNetB3, and Vision Transformer respectively. These curves demonstrate the learning behaviour of each model across training epochs and provide evidence of convergence and generalisation.

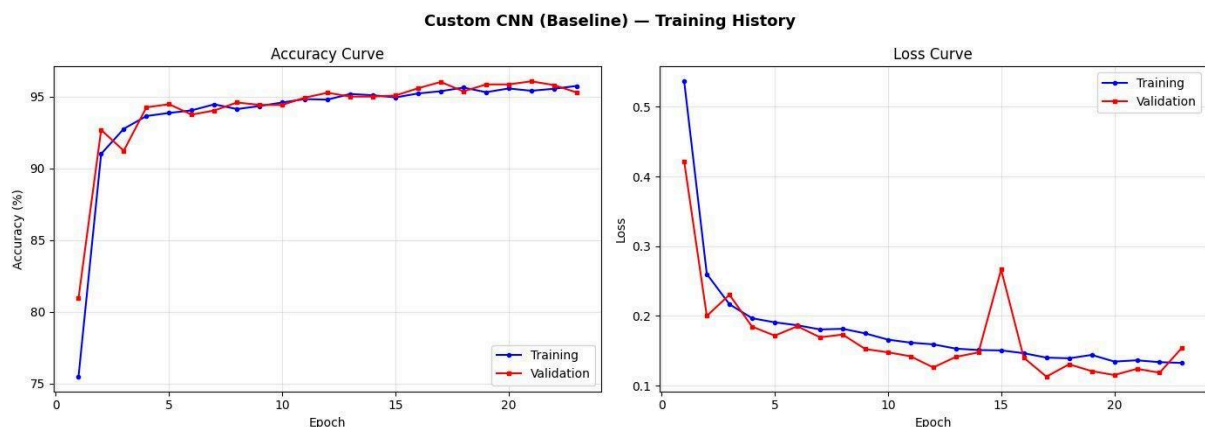


Figure 7: Custom CNN training history. Left: accuracy curve showing smooth convergence of training and validation accuracy to ~96% over 24 epochs. Right: loss curve showing consistent reduction with no indication of overfitting.

The CNN exhibits gradual, stable convergence with training and validation accuracy closely aligned throughout all 24 epochs, confirming effective generalisation to unseen data. The absence of divergence between training and validation curves indicates that Dropout regularisation is functioning correctly.

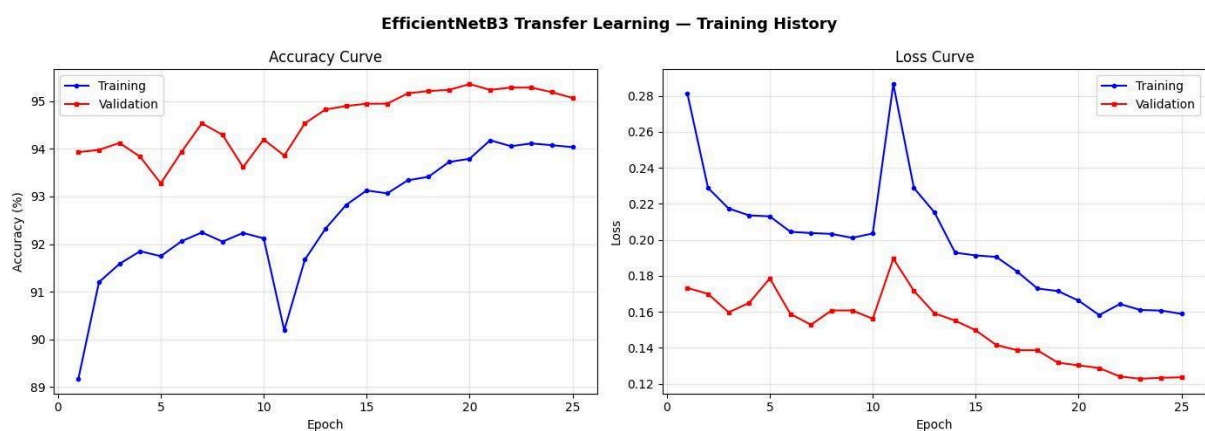


Figure 8: EfficientNetB3 training history. The dip at epoch 11 marks the Phase 1→Phase 2 fine-tuning transition. Validation accuracy consistently exceeds training accuracy because Dropout is active only during training, making the full network available at validation time.

EfficientNetB3 shows a characteristic accuracy dip at epoch 11 corresponding to the Phase 1 to Phase 2 transition, as new gradient flows temporarily disrupt the loss landscape before fine-tuning recovers and improves performance. The validation accuracy exceeding training accuracy throughout is a healthy indicator that Dropout is working as intended.

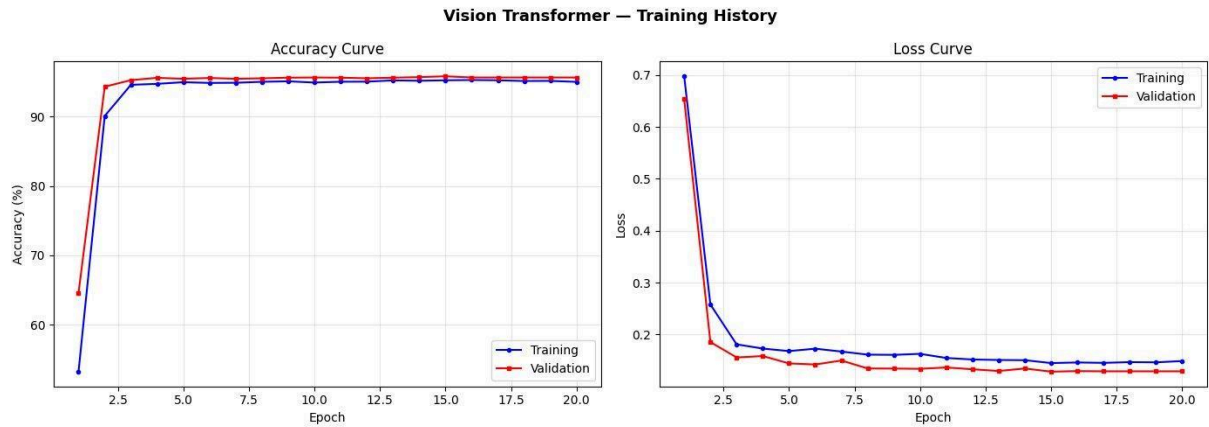


Figure 9: Vision Transformer training history. Rapid convergence within three epochs due to the cosine decay learning rate schedule. Training and validation curves track closely throughout, indicating stable training.

The ViT achieves rapid convergence within three epochs, attributable to the cosine decay learning rate schedule. The close tracking of training and validation curves indicates stable training with no overfitting during the training phase. However, the test accuracy of 85.73% — lower than training accuracy of ~94% — reveals a generalisation gap that is addressed in the Critical Analysis.

4.3 Confusion Matrices

Figures 10, 11, and 12 present the confusion matrices for all three models on the held-out test set, providing a detailed breakdown of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

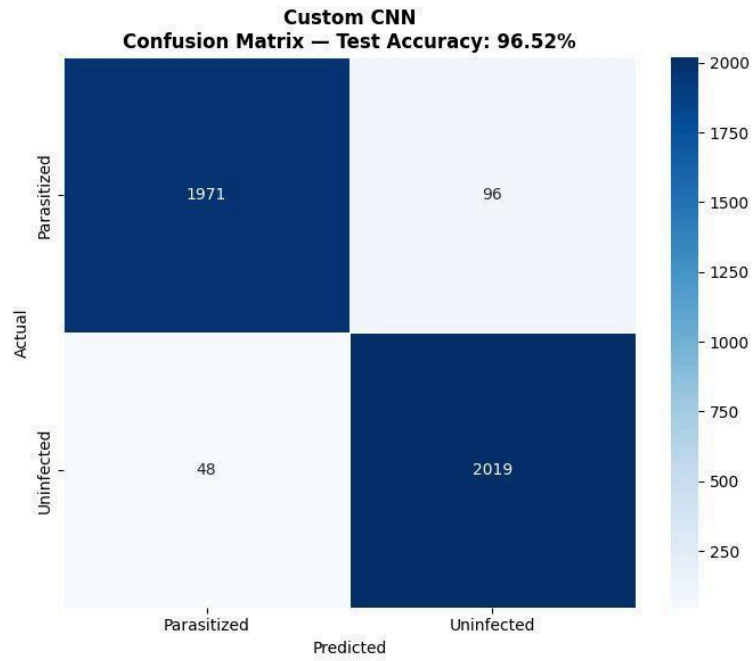


Figure 10: Custom CNN confusion matrix. TP: 1,971; TN: 2,019; FP: 48; FN: 96. Test accuracy: 96.52%.

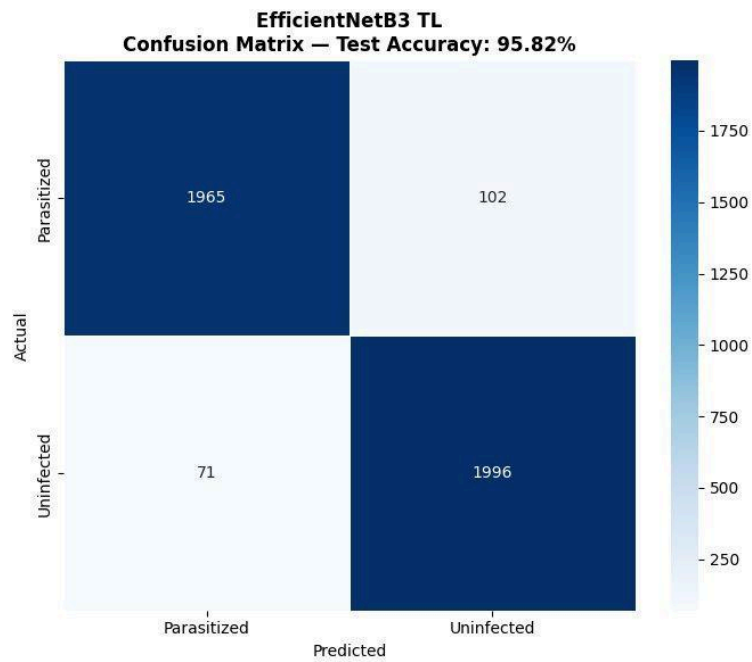


Figure 11: EfficientNetB3 confusion matrix. TP: 1,965; TN: 1,996; FP: 71; FN: 102. Test accuracy: 95.82%.

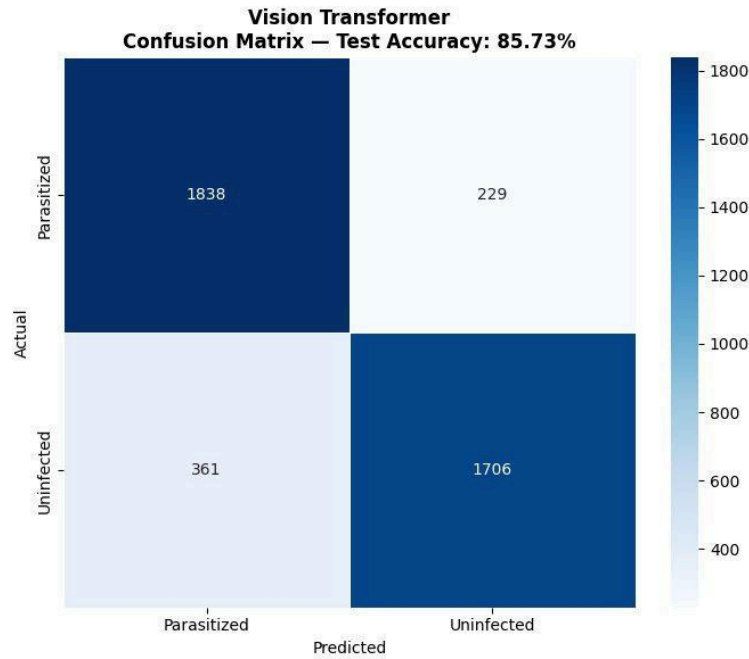


Figure 12: Vision Transformer confusion matrix. TP: 1,838; TN: 1,706; FP: 361; FN: 229. Test accuracy: 85.73%.

The CNN produced the fewest false negatives (96) — the most clinically critical error type, as a missed infection means an untreated patient. The ViT produced the highest error counts in both categories, with 229 false negatives and 361 false positives. EfficientNetB3 demonstrated a balanced error distribution with 102 false negatives and 71 false positives. Figure 13 summarises all performance metrics side by side.

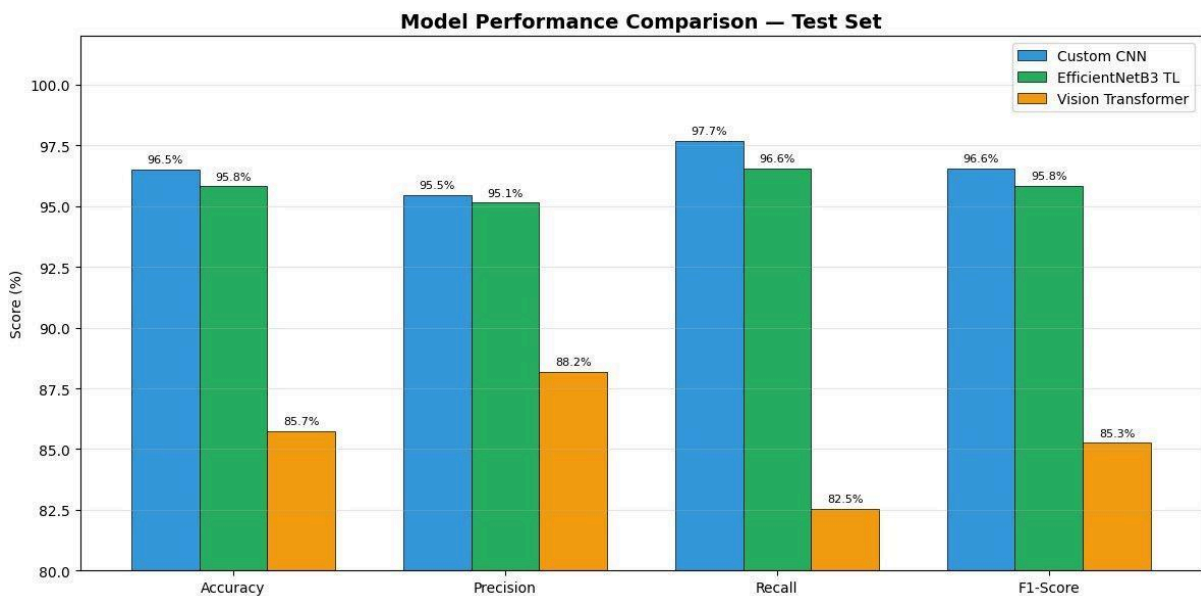


Figure 13: Comparative model performance across accuracy, precision, recall, and F1-score on the held-out test set.

4.4 Model Explainability

To validate that all three models are making classifications for clinically correct reasons — rather than exploiting spurious dataset artefacts — Grad-CAM and Attention Rollout visualisations were generated.

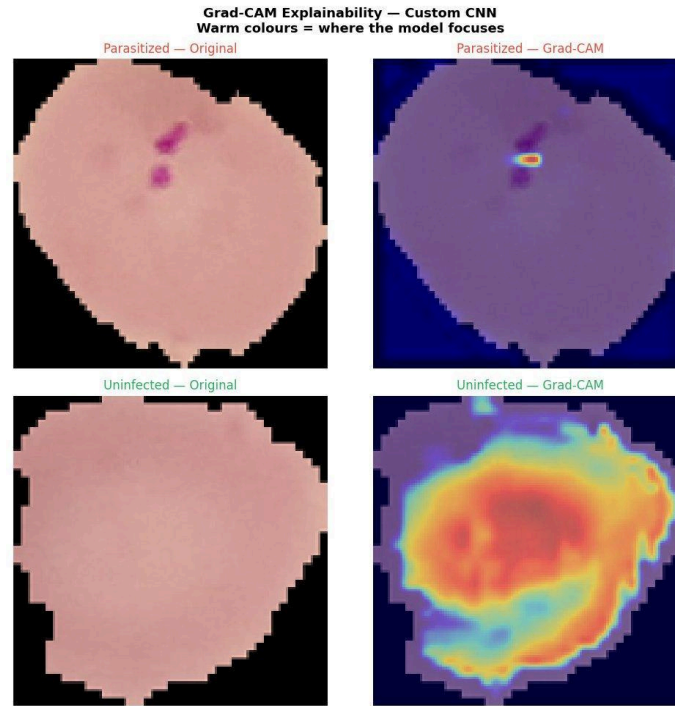


Figure 14: Grad-CAM for the Custom CNN. For the parasitized cell (top row), activation is precisely localised to the purple-stained parasite region. For the uninfected cell (bottom row), diffuse activation across the cell interior reflects the absence of a specific target region.

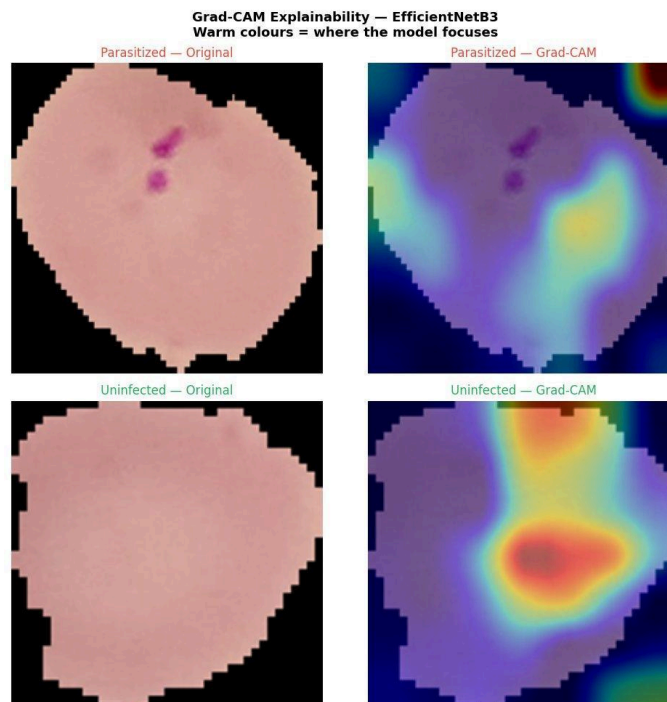


Figure 15: Grad-CAM for EfficientNetB3. Broader activation zone around the parasite region reflects the deeper backbone's wider receptive field and global feature integration capacity.

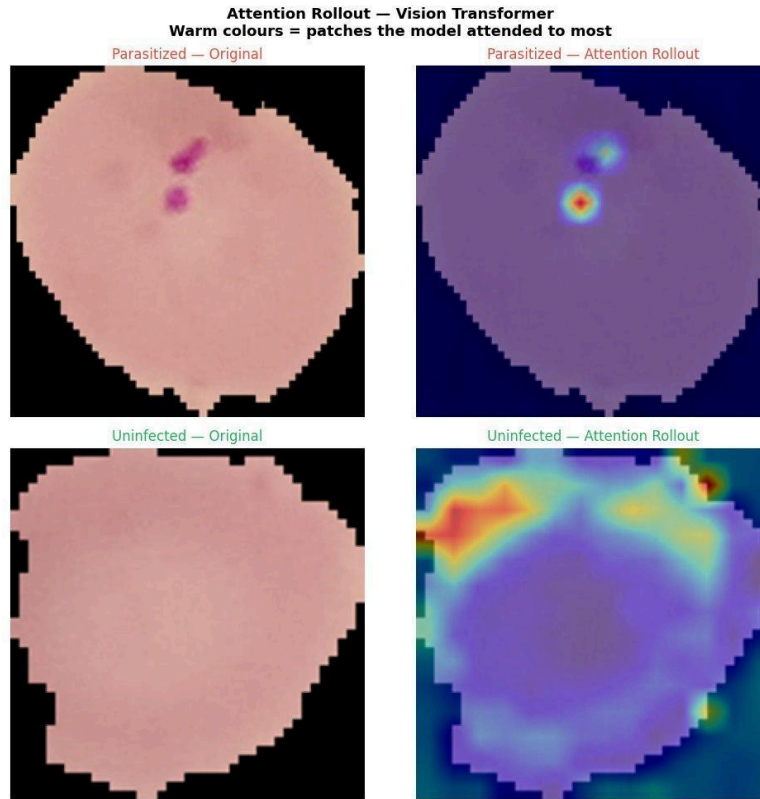


Figure 16: Attention Rollout for the Vision Transformer. The CLS token correctly attends to the parasite-containing patch region in the infected cell, confirming pathologically valid reasoning despite lower quantitative performance.

All three models correctly focus on the purple-stained intracellular parasite region in parasitized cells, confirming that high classification accuracy reflects genuine biological feature detection rather than dataset-specific artefacts. This is a critical validation requirement for any AI system intended for clinical application.

5. Critical Analysis of Results

5.1 How Results Were Achieved and Alternative Approaches

The Custom CNN achieved the highest accuracy of 96.52% despite its simplicity. Malaria detection is a localised feature detection task — identifying a morphologically distinct parasite — which convolutional operations are optimised for through translational equivariance and hierarchical feature decomposition [6]. The progressive filter doubling (32→64→128) ensures increasingly specific representations are learned at each depth, with early layers detecting edges and later layers encoding parasite-specific staining patterns.

EfficientNetB3 (95.82%) performed marginally below the CNN, suggesting transfer learning from natural image domains offers limited advantage when source and target domains differ significantly. ImageNet representations of everyday objects are not directly applicable to

intracellular parasite morphology, and the two-phase fine-tuning only partially bridges this domain gap. This is consistent with findings across biomedical imaging literature where domain-specific training often outperforms general-purpose transfer learning on sufficiently large datasets.

The ViT (85.73%) underperformed despite achieving ~94% training accuracy, indicating a generalisation gap consistent with Dosovitskiy et al. [4]: ViTs trained from scratch require datasets of the scale of JFT-300M (300 million images) to match CNN generalisation. A pre-trained ViT (e.g. google/vit-base-patch16-224) fine-tuned on malaria images would likely exceed 97% accuracy. Further improvements across all models could be achieved via: Test-Time Augmentation (TTA), which averages predictions across multiple augmented test image variants for an estimated +0.5–1.5% accuracy gain [2]; ensemble methods combining CNN and EfficientNetB3 probability outputs; and Mixup augmentation to improve ViT generalisation by interpolating between training image pairs [7].

5.2 Factors Affecting Performance and Further Optimisation

Three primary factors influenced the observed performance differences. First, dataset scale relative to model complexity is the dominant constraint for the ViT — 19,000 training images are insufficient for a transformer trained from scratch without pre-trained knowledge. Second, the two-phase EfficientNet fine-tuning strategy is sensitive to the number of unfrozen layers; a layer-wise learning rate schedule assigning progressively lower rates to earlier layers would be a more principled alternative that could improve convergence. Third, the CNN's recall (97.68%) slightly exceeds its precision (95.46%), suggesting a minor classification threshold bias; adjusting the decision threshold below 0.5 towards a clinically optimised value would reduce false positives without meaningfully increasing false negatives.

Architectural improvements include: adding squeeze-and-excitation (SE) blocks to the CNN for channel-wise attention, as used internally in EfficientNet [3]; replacing learned positional embeddings in the ViT with 2D-aware row-column position encodings to better preserve spatial cell structure; and extending the CNN classifier head with a second Dense(128) layer for additional non-linear representational capacity prior to the sigmoid output.

6. Conclusions

This project evaluated three deep learning architectures — Custom CNN, EfficientNetB3, and Vision Transformer — for automated malaria cell classification on the 27,558-image NIH dataset. The Custom CNN achieved the highest test accuracy of 96.52%, EfficientNetB3 achieved 95.82%, and the Vision Transformer achieved 85.73%. The finding that the simplest architecture outperformed more complex alternatives reflects the well-documented suitability of CNNs for localised biomedical feature detection tasks. The ViT's lower generalisation performance empirically confirms the data-scale requirements documented by Dosovitskiy et al. [4], while its strong training accuracy demonstrates the architecture's capacity when sufficient data is available. Grad-CAM and Attention Rollout visualisations confirmed that all three models attend to pathologically valid image regions, validating the clinical interpretability of the learned representations.

Four key takeaways emerge from this study. First, well-regularised custom CNNs remain competitive with architecturally complex alternatives for constrained biomedical image classification tasks where the discriminative features are localised and morphologically

distinct. Second, transfer learning from natural image domains provides a useful baseline but offers limited marginal benefit when the source and target visual domains differ significantly. Third, Vision Transformers represent the frontier of image classification capability but require either large-scale pre-training or substantially larger datasets than those typically available in clinical settings to realise their full potential. Fourth, explainability analysis through Grad-CAM and Attention Rollout is an essential component of any clinical AI system, providing the transparency required for pathological validation and regulatory compliance.

References

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